Feature Scaling Approaches :-

Z-score standardization (standardization/ scaling)

→ based on the standard normal distribution.

→ Z-score standardization transforms data such that it has $\boxed{\text{mean} = 0}$ and standard deviation = 1 : $\boxed{SD = 1}$

$$Z = \frac{x - \mu}{\sigma}$$

x = value $\quad \sigma$ = standard

μ = mean of data $\quad$ deviation

→ TO compare value from diffrent scales.

① (e.g. marks of 100 vs height in mm)

② one feature "age(year)' ≈ 20-60

another = "Salary ($) ≈ 30,000-100000

③ Raw values can't be compared, standardization puts both on the same scale (mean 0, SD=1)

→ Useful when the data may contain o
(Since z-score show how far a valu
is from the mean).

→ Needed for algorithms like Logistic
regression, KNN, PCA where variance matters.

Ex If a student score 90 marks in
the subject (mean = 70, sd = 10)

$$z = \frac{90-70}{100}$$

this means the student is 2 standard
deviation above the mean.

Important :-

(1) Helps in fair comparison where units
and scales are different.

(2) useful for outlier detection (e.g. fraud
detection, abnormal medical test results

(3) Needed in algorithms that assume normal
or depend on variance.

Case standardization (using sample SD)

dataset = [10, 12, 12, 13, 15, 18, 20, 22]

$n = 8$

step (1): - mean

$$\bar{x} = \frac{[10+12+12+13+15+18+20+22]}{8}$$

$$= \frac{122}{8} = 15.25$$

step (2) Deviation and squares (for varience)

$(x - \bar{x}) = [-5.25, -3.25, -3.25, -2.25, -0.25, 2.75, 4.75, 6.75]$

Squals = [27.565, 10.5625, 10.5625, 5.0625, 0.0625, 7.5625, 22.5625, 45.625]

Sum of squared deviations = 129.5

Sample of varience & SD

$$(\text{variance})\ S^2 = \frac{129.5}{n-1} = \frac{129.5}{8-1} = \frac{129.5}{7} = 18.5$$

$$(\text{SD})\ S = \sqrt{18.5} = 4.3012$$

Step ④ Z-Scores

$$z_i = \frac{x_i - \bar{x}}{s} = \frac{\text{data} - \text{mean}}{\text{SD}(S)}$$

data	z_i
10	$-5.25/4.3012 = -1.2206$
12	$-3.25/4.3012 = -0.7556$
12	$-3.25/4.3012 = -0.7556$
13	$-2.25/4.3012 = -0.5231$
15	$-0.25/4.3012 = -0.0581$
18	$2.75/4.3012 = 0.6394$
20	$4.75/4.3012 = 1.1044$
22	$6.75/4.3012 = 1.5693$

Check Z-score have mean ≈ 0 and SD = 1.

Interpretation Exp.: A score of 20 has $z \approx 1.10$, it mean 1.1 SD above the class average.

= 18.5

what does it mean when standardize data has mean = 0 and standard deviation = 1.

When we standardize (z-score transform)

$$Z = \frac{x - \mu}{\sigma}$$

After transformation.

- mean (center) becomes 0
- standard deviation (spread) becomes 1

this means :-

1. Data is now centred around 0 instead of the original mean.
2. Data spread is adjusted. so that its variability is measured in units of standard deviation.

Ex :- suppose we have a dataset

$$X = \{2, 4, 6, 8\}$$

mean $\mu = \frac{2+4+6+8}{4} = \frac{20}{4} = 5$

$$\sigma^2 = \frac{\sum (x_i - \mu)^2}{N} \quad , \sigma = \sqrt{\frac{(2-5)^2+(4-5)^2+(6-5)^2+(8-5)^2}{4}}$$

$$\sigma = \sqrt{5} = 2.236$$

Apply Z-score transformation :-

$$Z = \frac{x - \mu}{\sigma}$$

$2 \rightarrow (2-5)/2.236 = -1.34$

$4 \rightarrow (4-5)/2.236 = -0.45$

$6 \rightarrow (6-5)/2.236 = 0.45$

$8 \rightarrow (8-5)/2.236 = 1.34$

$Z = \{-1.34, -0.45, 0.45, 1.34\}$

check mean of z.

$$\mu_z = \frac{-1.34 + (-0.45) + 0.45 + 1.34}{4}$$

$$= 0$$

mean becomes 0.

check stand deviation of z

$$\sigma_z = 1.$$

standard deviation becomes = 1.

Key points :-

. standardization rescale the features such as

mean (μ) = 0 and standard deviation (σ) = 1.

$$z = \frac{x - \mu}{\sigma}$$

. the result of standardization is z called.

as z-score normalization.

* if data follow normal distribution (gaussian distributions).

* if the original distribution is normal, then the standardised distribution will be normal.

* if the original distribution is skewed, then the standardised distribution of the variable will also be skewed.

What is Normalization = ?

→ Normalization rescale the features in fixed range between range 0 to 1.

$$X_{norm} = \frac{X - X_{min}}{X_{max} - X_{min}}$$

→ Normalization also called as min-max scaling

→ if data doesn't follow normal distribution (Gaussian)

Standardization vs Normalization ??

→ But mostly standardization use for clustering analysis, principal Component Analysis. (PCA).

→ Normalization prefer for image processing bcz pixel intensity b/w 0 to 255, neural network algorithm require data in scale 0-1,

Impact

When we work with machine learning algorithms, features (variable) have different range (e.g. age = 1–100, income 100000 – 10000000.

- features with large numerical values dominate model like KNN, SVM, Gradient descent.

that's why we use scaling techniques like standardization and normalization.

① Standardization (Z-score normalization)

$$Z = \frac{x - \mu}{\sigma}$$

x = original data, μ = mean, σ = SD.

effect

→ Transforms data to zero mean and unit variance.

→ values are centered typically within range [-3, +3]

When to use

→ When data has outliers
→ When algorithm assume gaussian distribution & linear regression, Logistic Reg, SVM, PCA.

Zero mean

After standardization, the avg(mean) of the transformed data becomes 0.

what means, if you add up all the scale value and divide by the number of data points → you will exactly 0

so the data is shifted to centred around 0.

Unit variance :-

variance (spread of data) after scaling becomes SD = 1 means; on average, most data point are about 1 unit away from the mean (0)

* Standardize score have SD = 1,
unlike the original data values

Std dev = 1 & 7

so the scale is now consistent
across all feature.

step ③ typically within in [-3, +3]

in statistics, if data follows a normal
distribution (bell curve)

about 68% of values lies b/w -1 and +1
—— 95% —— -2 and +2
—— 99.7 —— -3 and +3

this is called empirical rule (68-95-99.7 rule)

so, after standardization, most of the
values (if they are normally distributed) will
fall in range [-3, +3].

Standardization (Z-score scaling)

(1) When features are assumed to follow a Gaussian distribution.

(2) In models that assume data is centred around 0, e.g. linear regression, SVM, PCA.

(3) Works better with outlier bcz it spreads data based on standard deviation

(4) Output range is not fixed. (typically value fall b/w -3 and +3.)

Normalization (min-max scaling)

(1) When you need strictly in a fixed [0,1] or [-1,1]

(2) In model that rely on distance or similarity KNN, Neural network, clustering

(3) Sensitive to outlier bcz min or max changes can distort the scaling

(4) Output range is fixed ([0,1] or [-1,1].)

(5) Used in k-means, image processing, deep learning model.

Why Normalization is sensitive to outlier :-

$$x' = \frac{x - \min(x)}{\max(x) - \min(x)}$$

Scaling depends directly on minm and maximum value.

② if even one extreme outlier appears the min or max changes drastically. Then, all the other "normal" data gets squished into a tiny range.

Ex① with outlier.

D = [10, 12, 14, 15, 18]

min = 10, max = 18

Normalized value [0, 0.25, 0.5, 0.625, 1]

Clean scaling, value nicely spread b/w 0-1

eg ② with an outlier

$D = [10, 12, 14, 15, 18, 1000]$

$min = 10, \quad max = 1000$

Normalized value.

$[0, 0.002, 0.004, 0.005, 0.008, 1]$

Now see the problem

almost all value (10-18) are squashed near 0 (0 to 0.008)

the outlier 1000 dominates the scaling, stretching the whole range

Conclusion :- (Key points):-

Z score, Harmful standard deviation away a datapoint is from mean.

Exp Apply feature scaling

$X = [50, 5.2, 53, 5.4, 55, 57, 58,$
$60, 61, 62, 64, 65, 67, 68, 70]$

Calculate mean (μ)

$$= \frac{886}{15} = 59.07$$

Calculate mean standard deviation (σ)

$$= \sqrt{\frac{\sum (x_i - \mu)^2}{N}} = \sqrt{\frac{617.07}{15}}$$

$$= \sqrt{41.14} \simeq 6.41$$

Compute Z-Score

$$z_i = \frac{x_i - \mu}{\sigma}$$

$$z_i = \frac{x_i - 59.07}{6.41}$$